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1.1.3 What you will learn

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Numerical analysis is the study of how the perturbation of a problem or data affects the accuracy of computation. This inherently means that you have to be able to measure whether changes are large or small. That, in turn, means we need to be able to quantify whether vectors or matrices are large or small. Norms are a tool for measuring magnitude.

Upon completion of this week, you should be able to

- Prove or disprove that a function is a norm.
- Connect linear transformations to matrices.
- Recognize, compute, and employ different measures of length, which differ and yet are equivalent.
- Exploit the benefits of examining vectors on the unit ball.
- Categorize different matrix norms based on their properties.
- Describe, in words and mathematically, how the condition number of a matrix affects how a relative change in the right-hand side can amplify into relative change in the solution of a linear system.
- Use norms to quantify the conditioning of solving linear systems.

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